

ELEC 451 Power Electronics

Lab 0: Introduction to Simulations and Experimental Instruments

Christian Conroy, 24689630, Labib Kowsar, 89102628, Didier Gil Martinez, 43783323

I. INTRODUCTION

THIS is Lab 0 of ELEC 451. Its principal focus is on the basics of PSIM, the proper use and calibration of the measurement probes used in the lab (passive, differential and current probes), and performing mathematics such as the Fast Fourier Transform (FFT) in both PSIM and on the oscilloscope.

The overarching scientific objective of this lab, in addition to the practical skills, is to demonstrate how simulation and real-world results differ or overlap.

This report is organized as follows. Section II presents the PSIM simulations: circuit setup, transient and steady-state waveforms, the effect of the time step, mean-value measurements and the FFT. Section III describes the experimental work with the oscilloscope: probe calibration, time-domain measurements and the FFT. Section IV is a summary and conclusion of our findings.

II. SIMULATION

A. Circuit Setup (Activity 1)

The circuit in Fig. 1 was built in PSIM. A 10 kHz square-wave voltage source feeds a 1 mH inductor in series, followed by a 10 μ F capacitor in parallel with a 20 Ω resistor. The source voltage v_i , the output voltage v_o and the inductor current i_L are measured with probes placed according to the polarity shown.

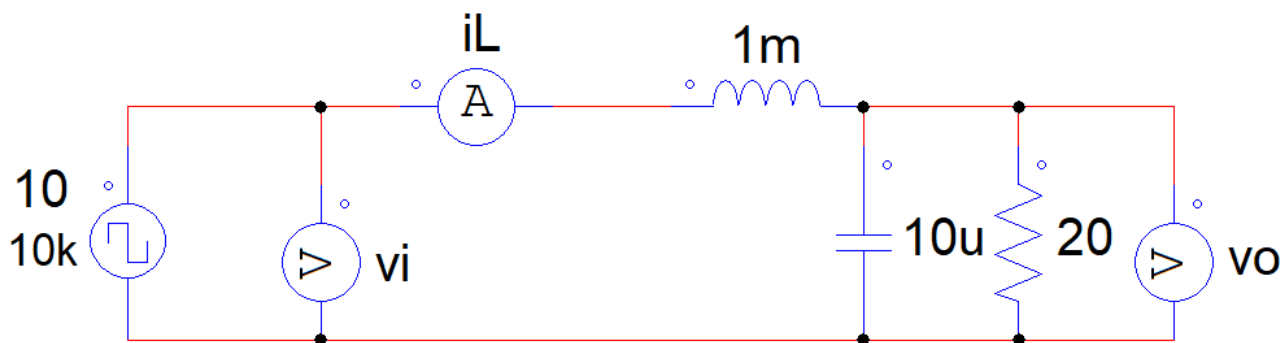


Fig. 1. PSIM schematic of the circuit: square-wave source (10 kHz), 1 mH inductor, 10 μ F capacitor and 20 Ω resistor, with voltage probes v_i and v_o and inductor current probe i_L .

B. Waveform Captures (Activities 2 and 3)

Fig. 2 shows v_i , v_o and i_L over the first 5.5 ms as first captured. The square-wave source correctly swings between 0 V and 10 V, as set on the source; v_o overshoots to about 7.3 V before settling to a steady-state ripple around 5 V.

The Simulation Control tool was then added, with a time step of 1 μ s and a total time of 2 ms, so that the complete transient of the signals is visible (Fig. 3); the signals reach steady state after roughly 1 ms. Fig. 4 zooms in to five cycles of the steady-state waveforms.

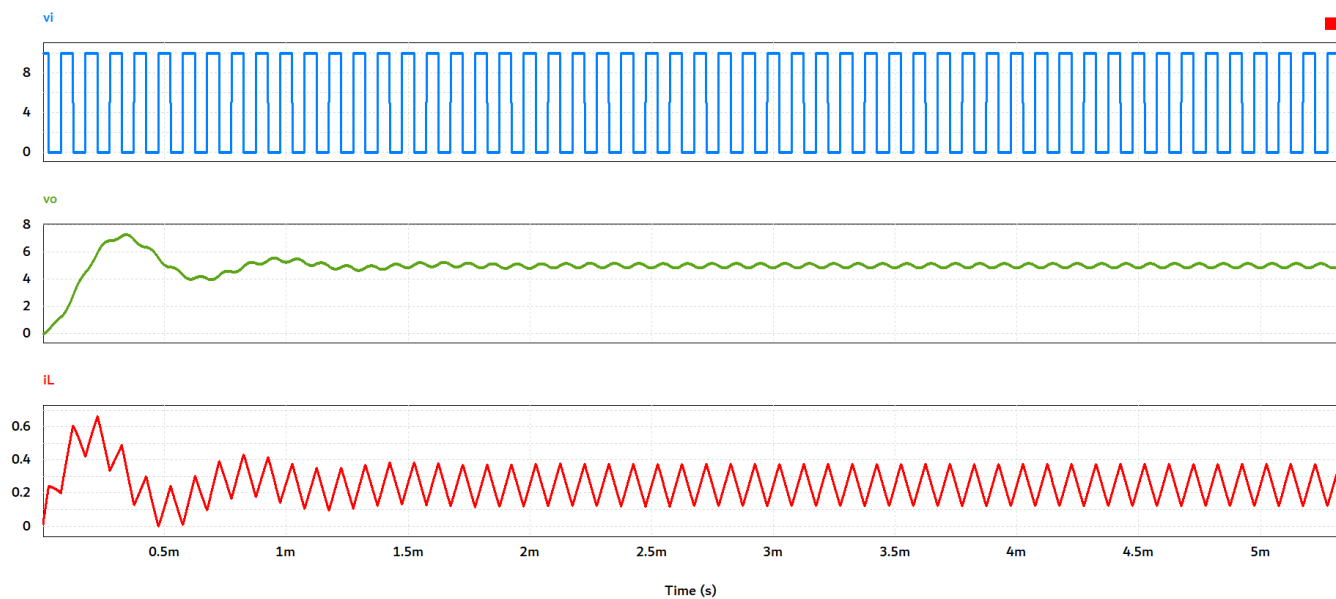


Fig. 2. Simulated waveforms v_i , v_o and i_L (top to bottom) as first captured.

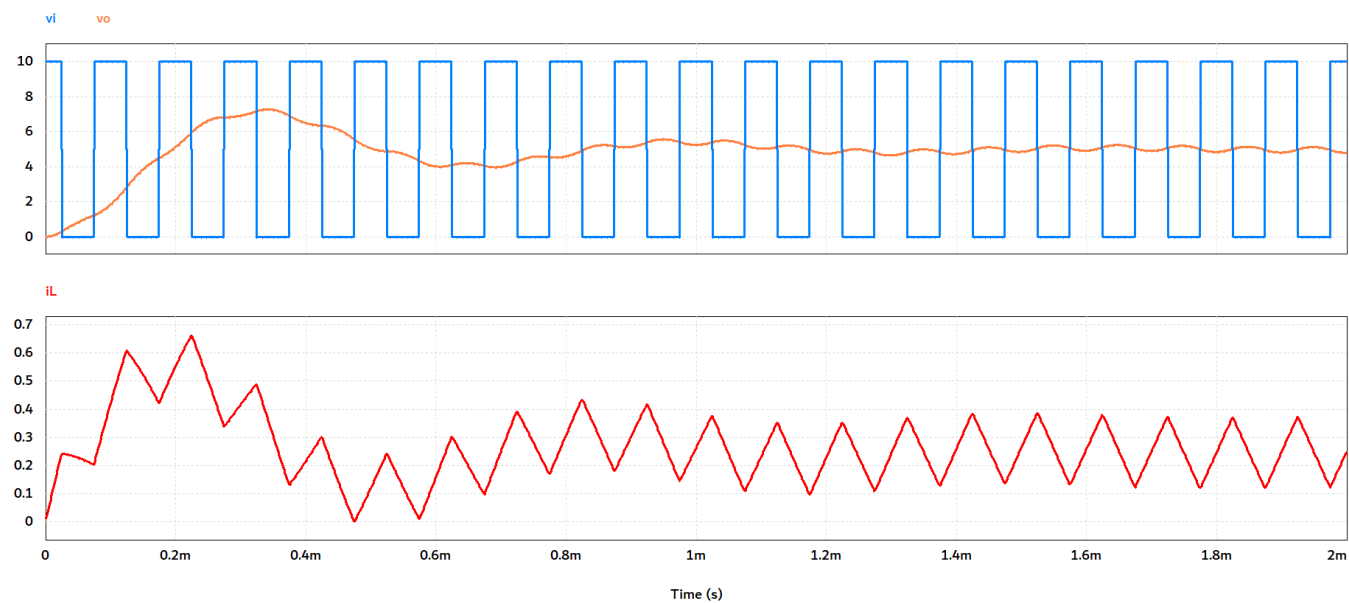


Fig. 3. Transient and steady-state waveforms with a $1\ \mu\text{s}$ time step and a total time of 2 ms: v_i and v_o overlaid (top), i_L (bottom).

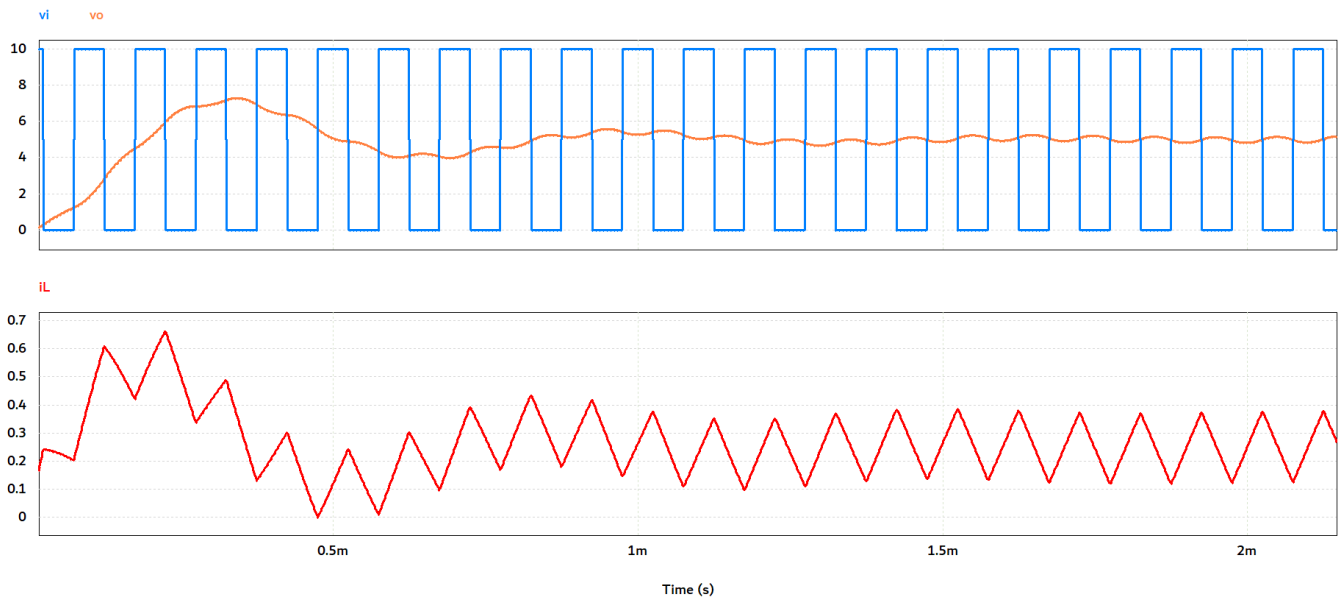


Fig. 4. Five cycles of the steady-state waveforms: v_i and v_o overlaid (top), i_L (bottom).

C. Time Step (Activity 4)

The waveforms were captured for a large time step (1×10^{-5} s, Fig. 5), a reasonable time step ($1 \mu\text{s}$, Fig. 6) and a very small time step (1×10^{-7} s, Fig. 7). Even at this scale, the waveforms remain clean, since 20Ω load strongly damps the ringing regardless of step size; a closer, few-cycle zoom would be needed to see any edge distortion at the largest step.

The more time steps taken, the longer the software takes to process the simulation.

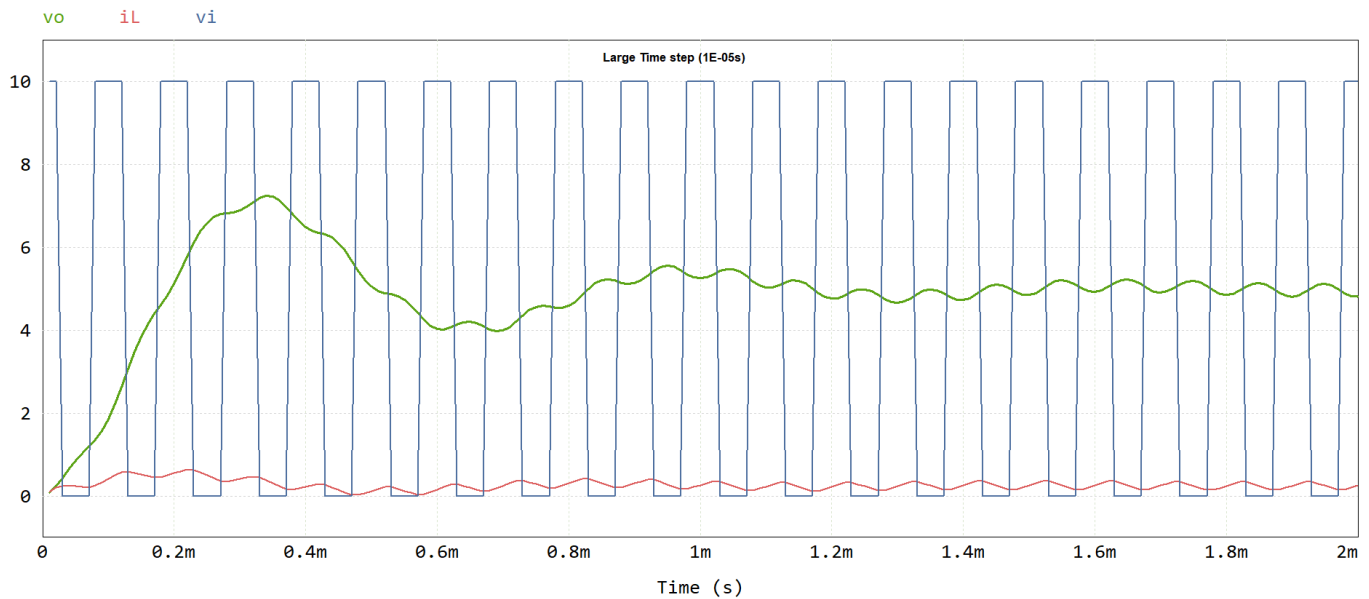


Fig. 5. Waveforms with a large time step (1×10^{-5} s): v_i , v_o and i_L overlaid.

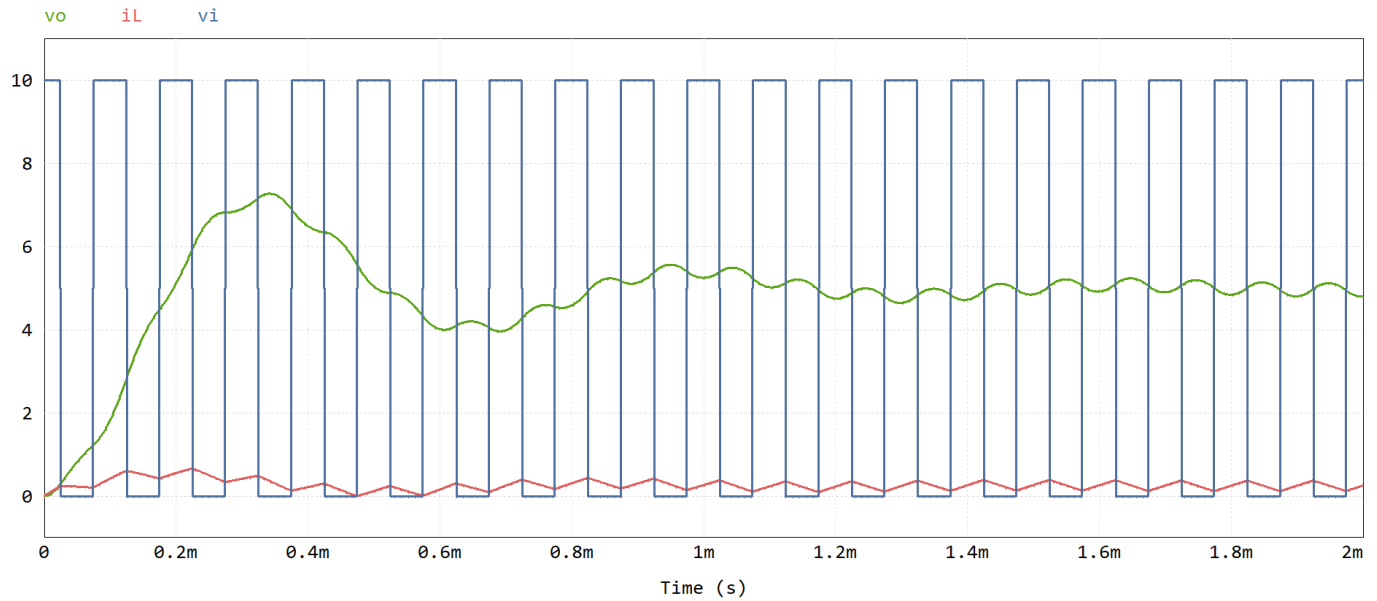


Fig. 6. Waveforms with a reasonable time step ($1 \mu\text{s}$): v_i , v_o and i_L overlaid.

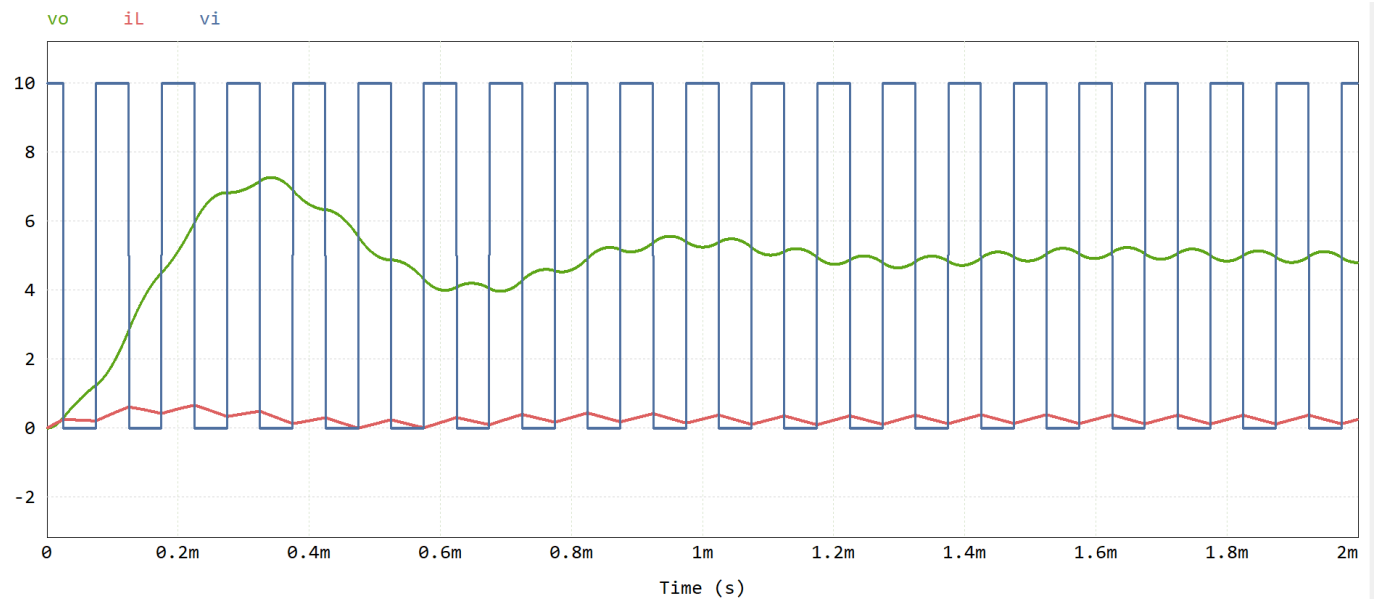


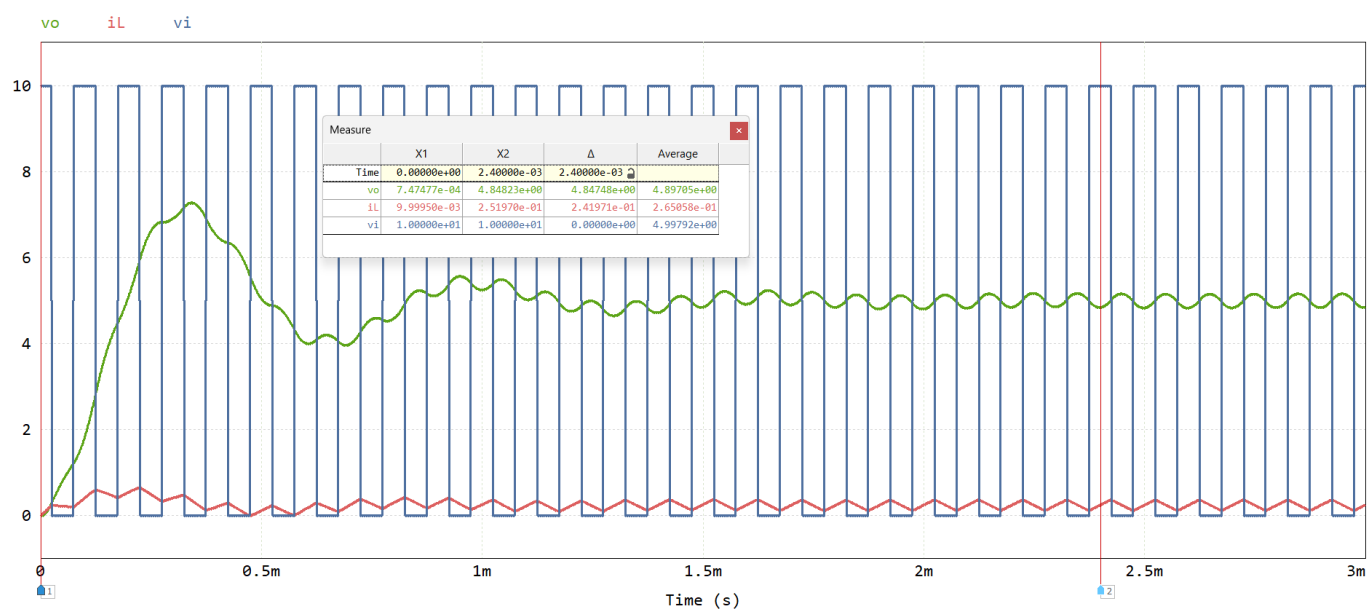
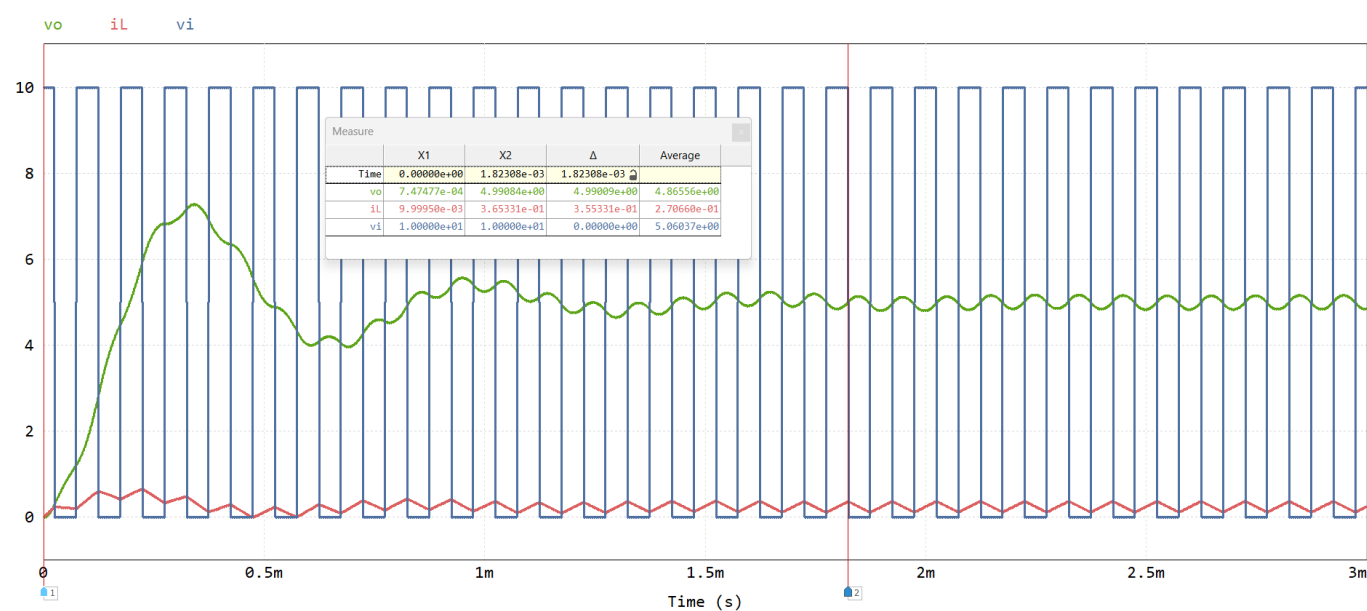
Fig. 7. Waveforms with a very small time step ($1 \times 10^{-7} \text{ s}$): v_i , v_o and i_L overlaid.

D. Measurements (Activity 5)

(a) **Mean values.** The mean values of v_o and i_L are shown in Fig. 8 for 10 cycles, Fig. 9 for 2 and a fraction of a third cycle, and Fig. 10 for several cycles (around 50). The results are compiled in Table I.

TABLE I
MEAN VALUES OF v_o AND i_L OVER DIFFERENT TIME SPANS

Time span	v_o (V)	i_L (A)
10 cycles	4.897	0.265
2.33 cycles	4.865	0.271
~ 50 cycles	4.975	0.254

Fig. 8. Mean value of v_o and i_L over 10 cycles.Fig. 9. Mean value of v_o and i_L over 2 and a fraction of a third cycle.

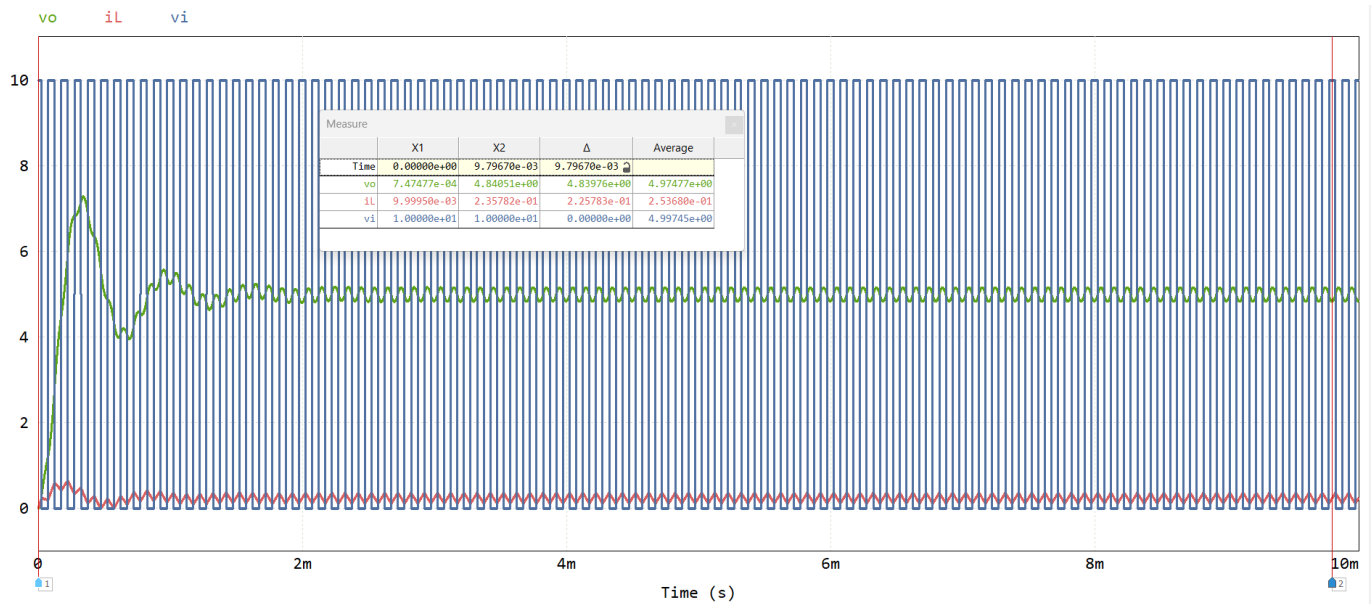


Fig. 10. Mean value of v_o and i_L over several cycles (around 50).

(b) Discussion. As the number of cycles increased, the v_o and i_L averages drifted towards their steady-state values of 5 V and 0.25 A, respectively. If allowed to average over infinite steady-state cycles, they would approach those values $\pm$ the peak of the ripple.

(c) Peak inductor current. The peak i_L current at 5 complete steady-state cycles is 0.379 A (Fig. 11).

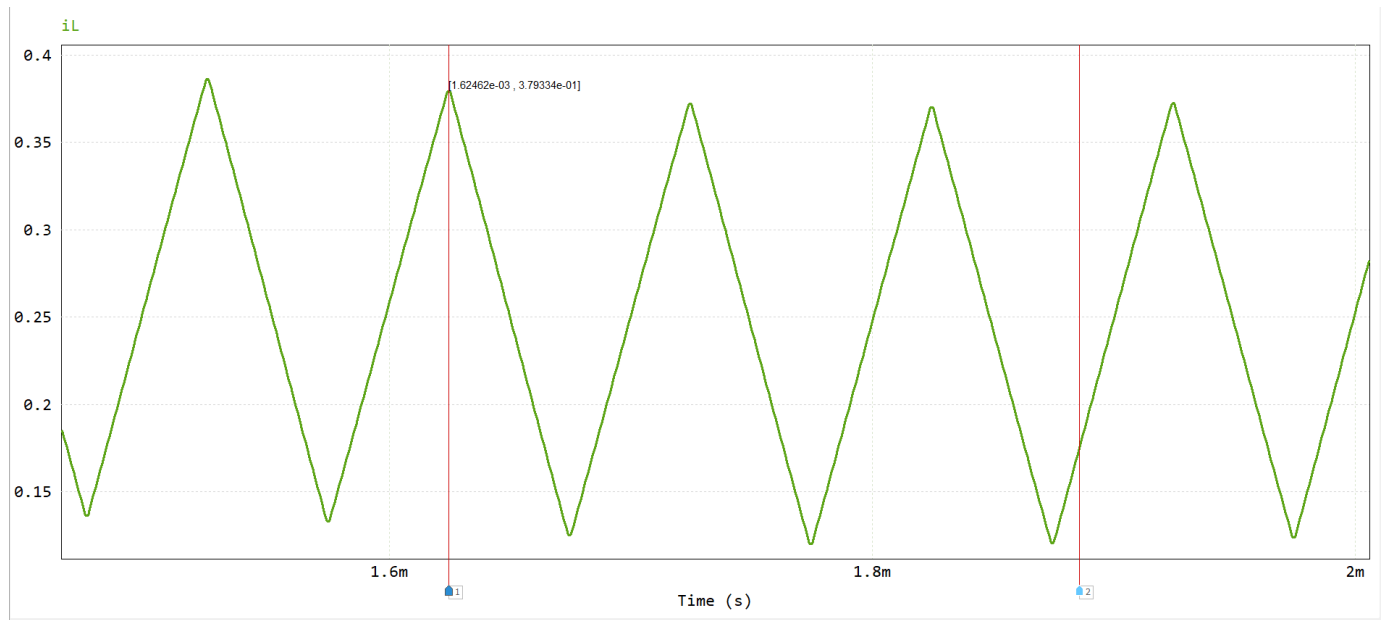


Fig. 11. Peak inductor current at 5 complete steady-state cycles (0.379 A).

(d) Inductor current waveform. The inductor-current-versus-time plot and its measurements are shown in Fig. 12.

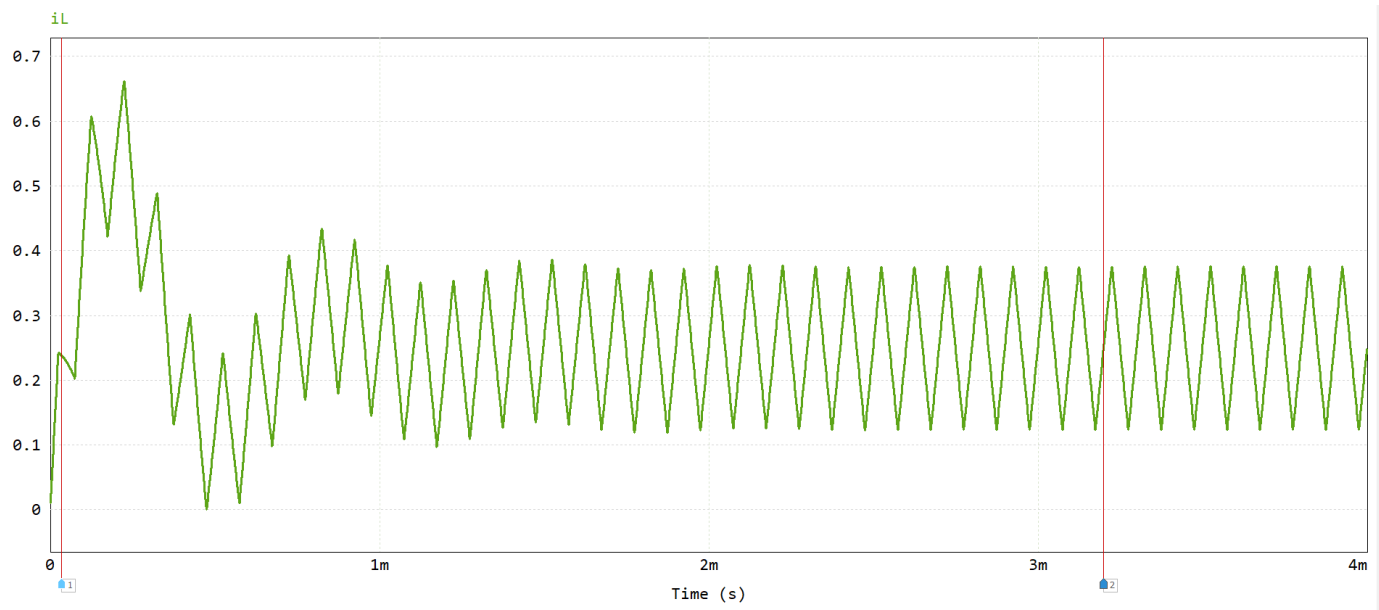


Fig. 12. Inductor current i_L versus time with cursor measurements.

E. FFT (Activity 6)

The circuit for this activity, shown in Fig. 13, is a 10 kHz triangular-wave voltage source (1 V peak-to-peak, 2 V DC offset) with a voltage probe measuring V_{tri} .

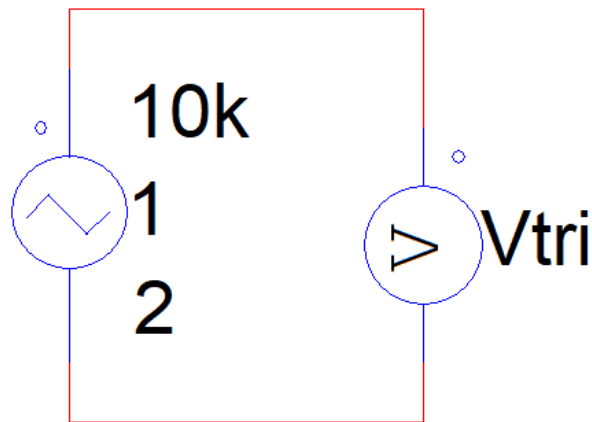


Fig. 13. PSIM schematic for Activity 6: 10 kHz triangular-wave voltage source and voltage probe V_{tri} .

Its output was simulated and its FFT computed with Analysis/Perform FFT (Fig. 14); the fundamental appears clearly at 10 kHz.

The total simulation time was then reduced to below 10 cycles (Fig. 15); the resulting FFT (Fig. 16) shows a wider, less well-defined peak at 10 kHz because of the coarser frequency resolution. With the total time increased instead (Fig. 17), the fundamental at 10.0 kHz and the third harmonic at 30.1 kHz are both sharply resolved.

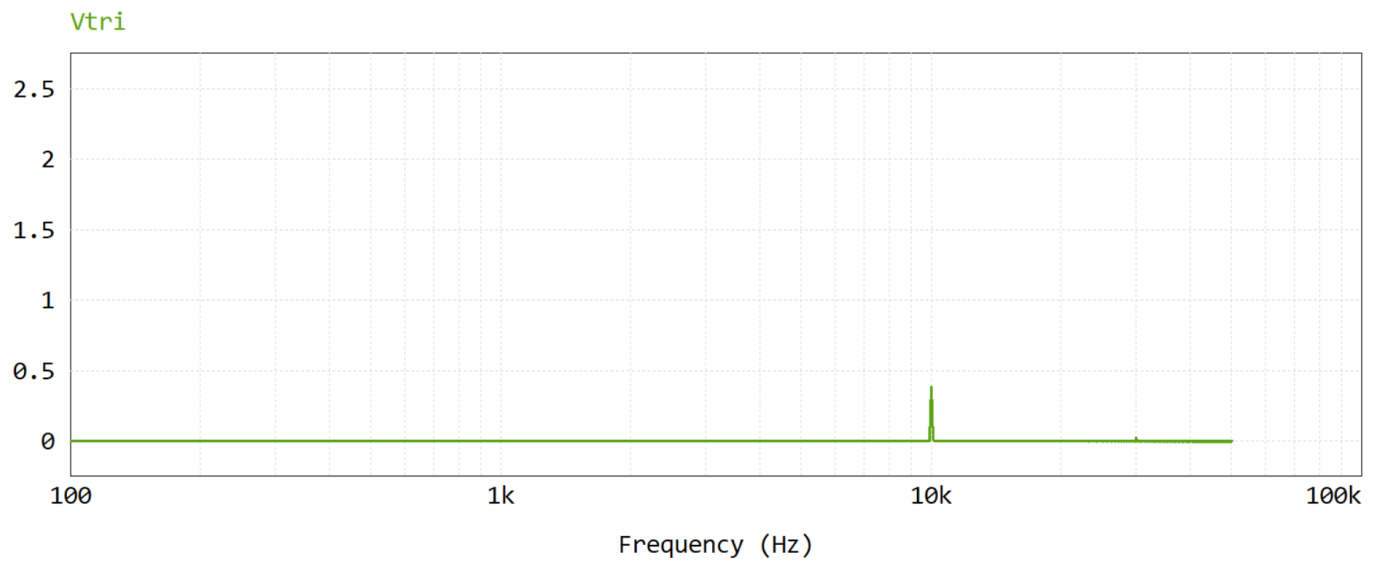


Fig. 14. FFT of the triangular-wave voltage, full span.

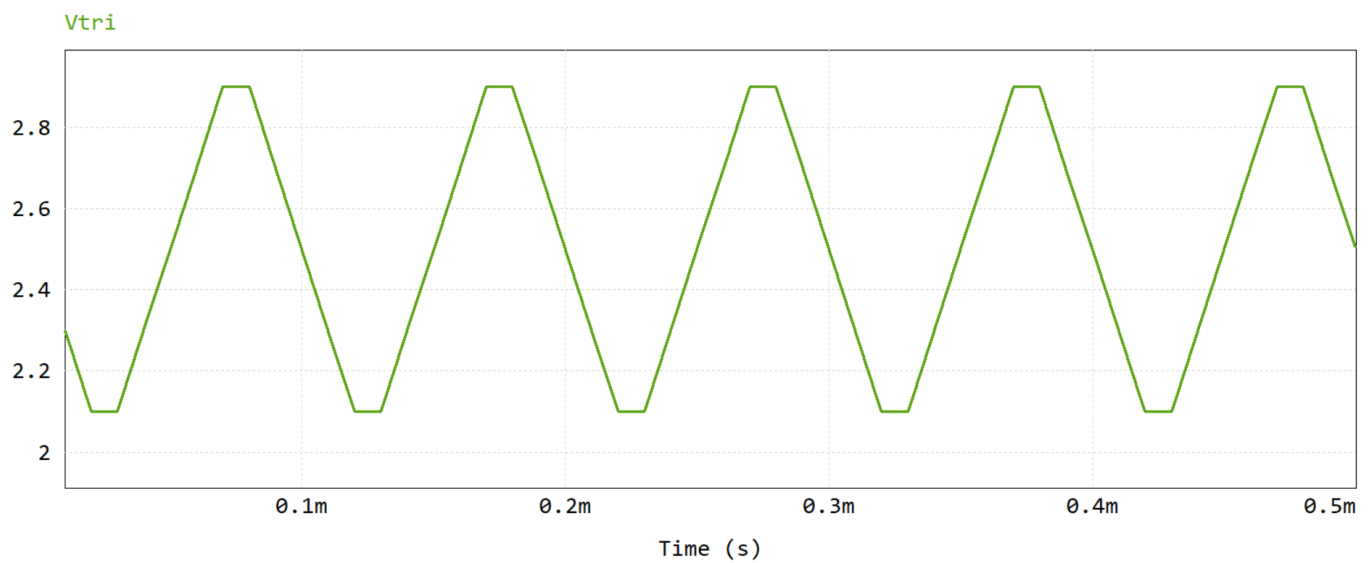


Fig. 15. Triangular-wave voltage with the total simulation time reduced to below 10 cycles.

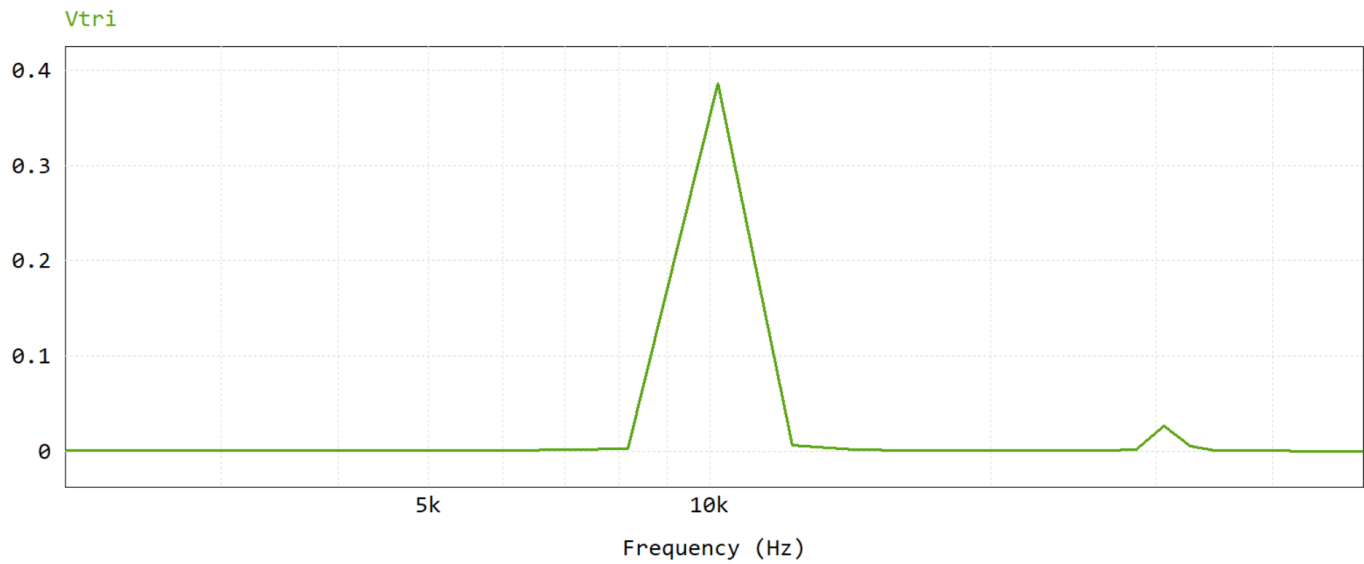


Fig. 16. FFT of the triangular-wave voltage with the total simulation time reduced to below 10 cycles.

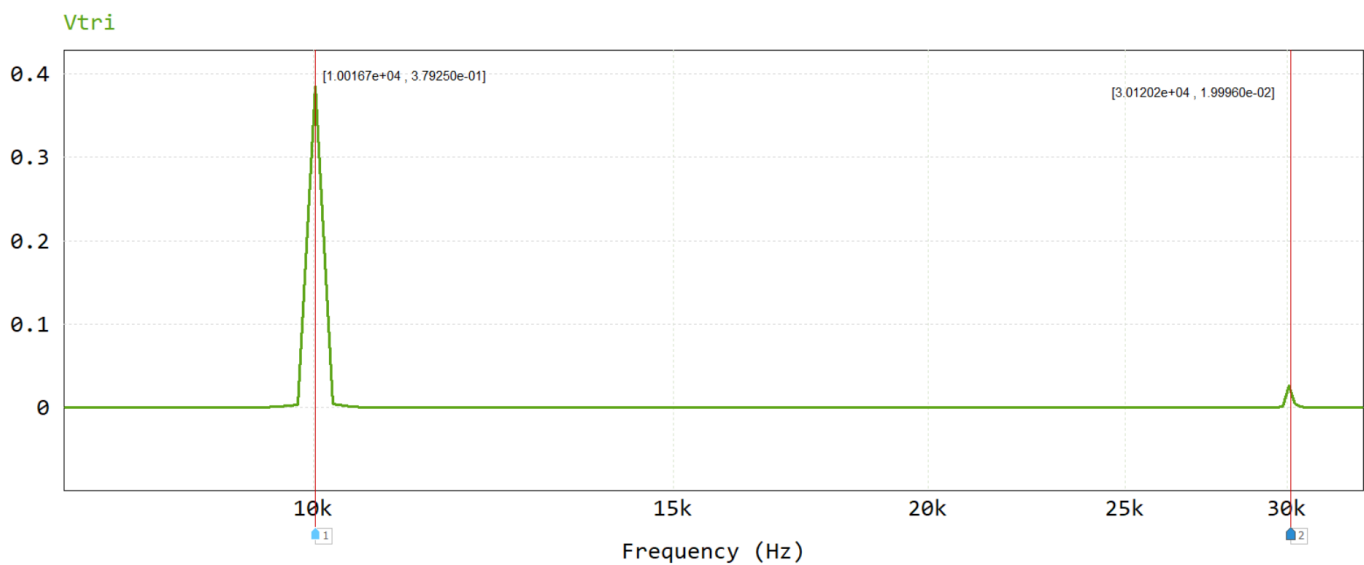


Fig. 17. FFT of the triangular-wave voltage with the total simulation time increased to more than 10 cycles. Cursors mark the fundamental (10.02 kHz, 0.379) and the third harmonic (30.12 kHz, 0.020).

A longer total simulation time gives finer frequency resolution and a sharper, more legible FFT, while a short total time smears the peaks and makes the harmonics harder to distinguish. This reinforces the same Shannon–Nyquist trade-off encountered when varying the time step: enough samples, over a long enough window, are needed to resolve the spectrum clearly.

III. EXPERIMENT

A. Probe Calibration (Activity 1)

Before taking measurements, the probes were configured and calibrated against the calibration signal of the oscilloscope (a 1 kHz square wave) (Fig. 18 and Fig. 19).

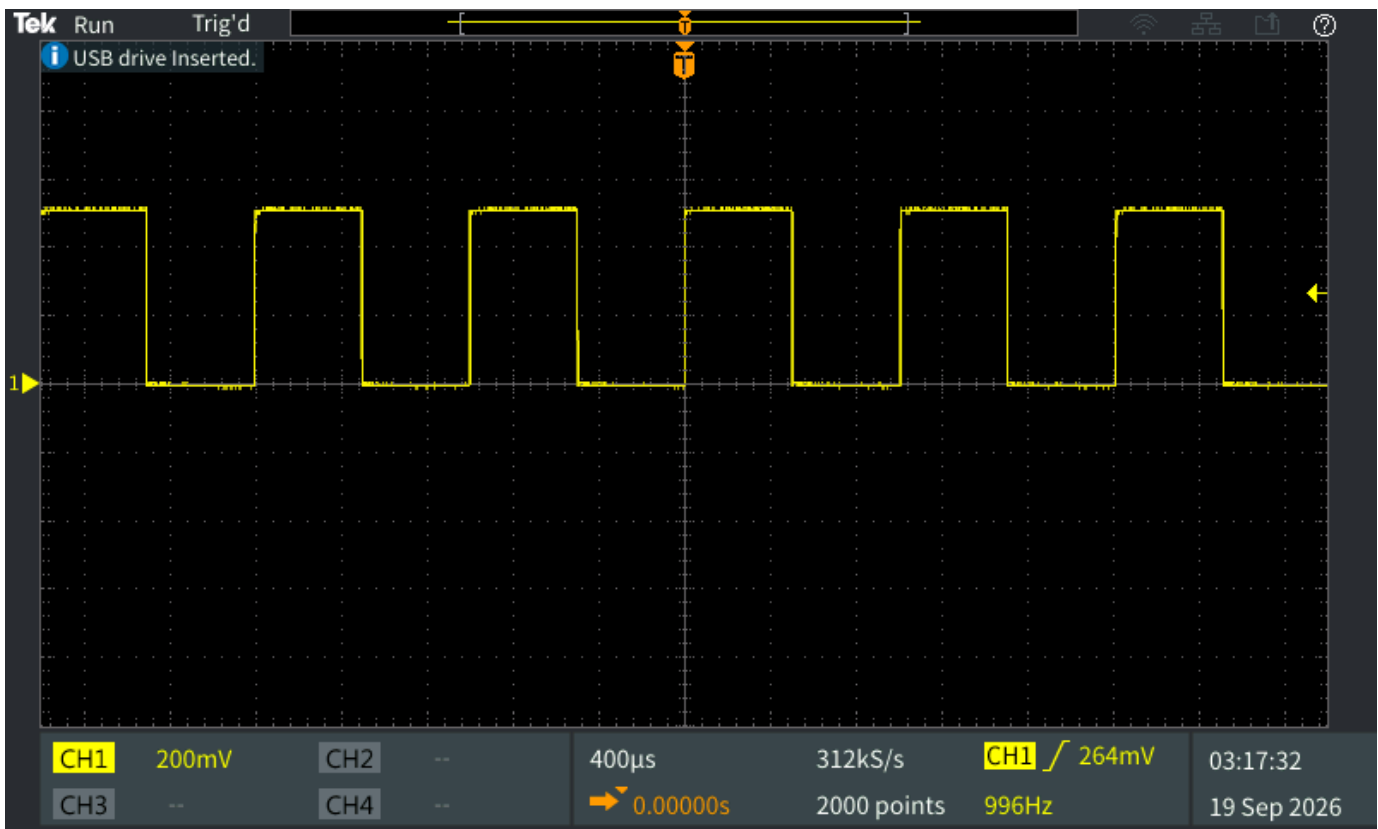


Fig. 18. Passive probe: oscilloscope capture of the calibration signal.

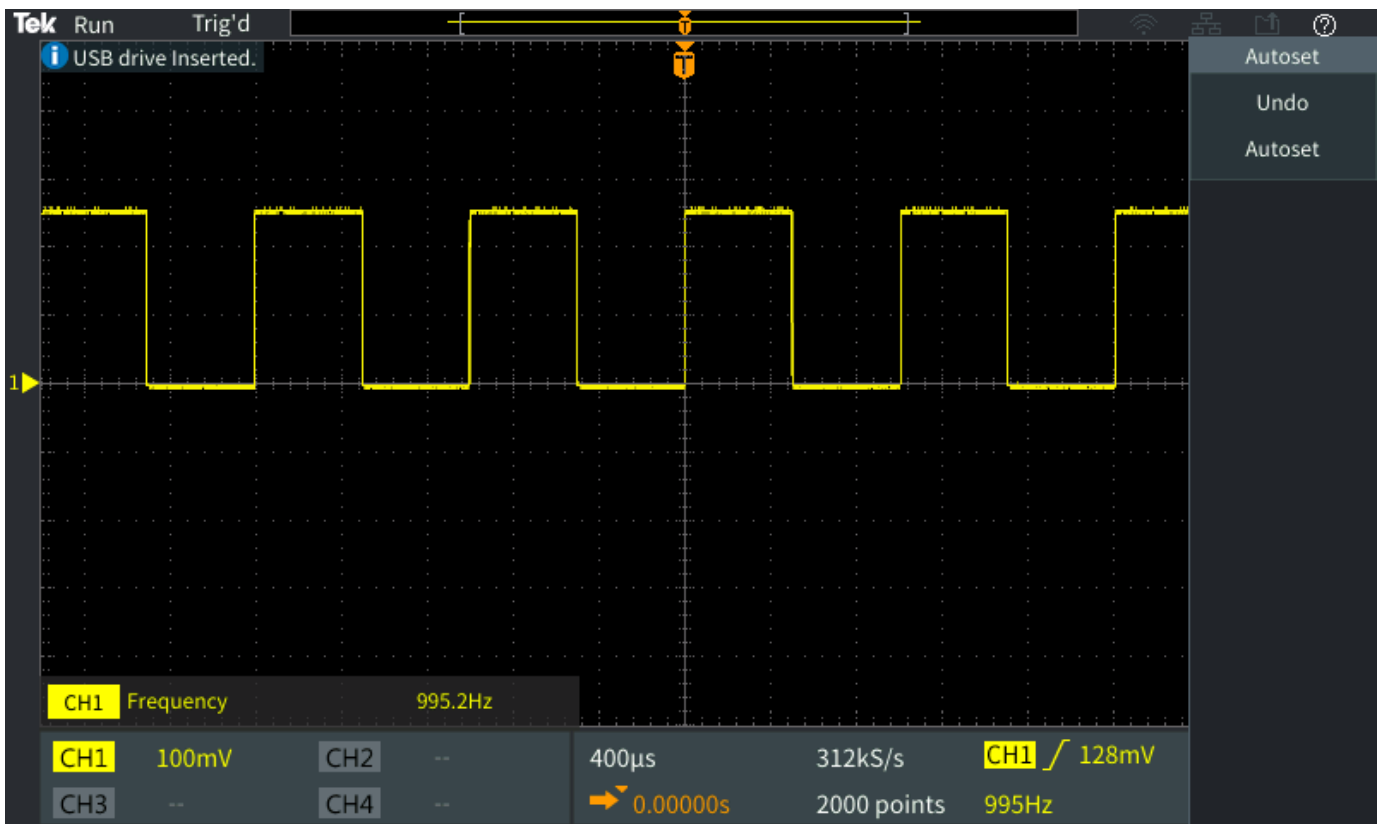


Fig. 19. Differential probe: oscilloscope capture of the calibration signal.

B. Time-Domain Captures and Measurements (Activity 2)

A triangular wave (10 kHz, 500 mV amplitude, 2 V DC offset) from the function generator was displayed on the oscilloscope (Fig. 20). The frequency and mean value measured by the scope (10.00 kHz and 1.995 V) match the generator settings, and the peak-to-peak amplitude read 560 mV for the 500 mV set on the generator.

When the amplitude was reduced to 10 mV, the details inside the signal became invisible because of noise. After changing the channel from DC to AC coupling and applying a high-frequency filter, the triangular shape could be recovered (Fig. 21): the ripple came into view and, although the signal was still noisy, it could be analyzed and understood. The automatic measurements were nevertheless still affected by the noise, with 15.6 mV peak-to-peak and 17.72 kHz reported for a 10 mV, 10 kHz signal.

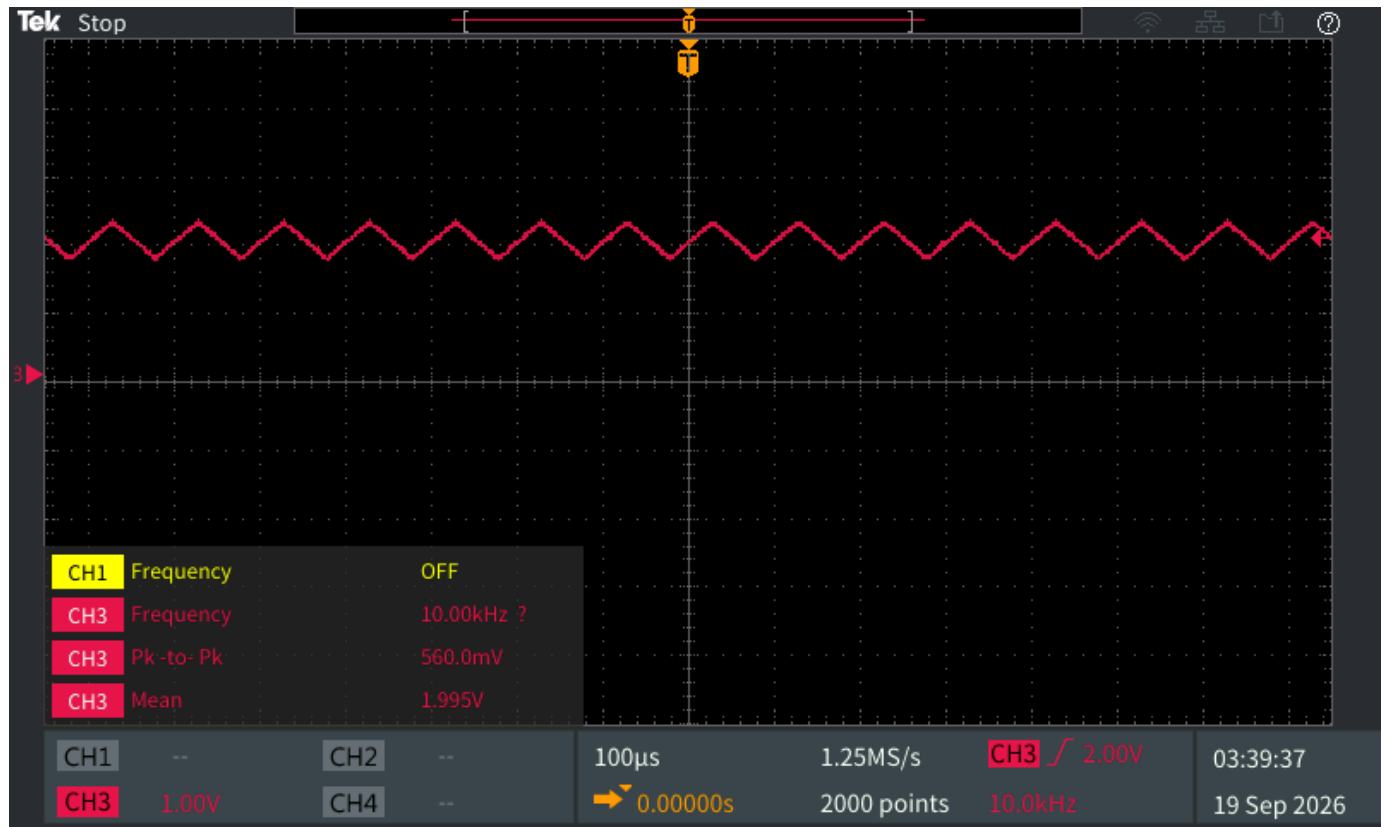


Fig. 20. Oscilloscope capture of the 10 kHz triangular wave with 500 mV amplitude and 2 V offset.

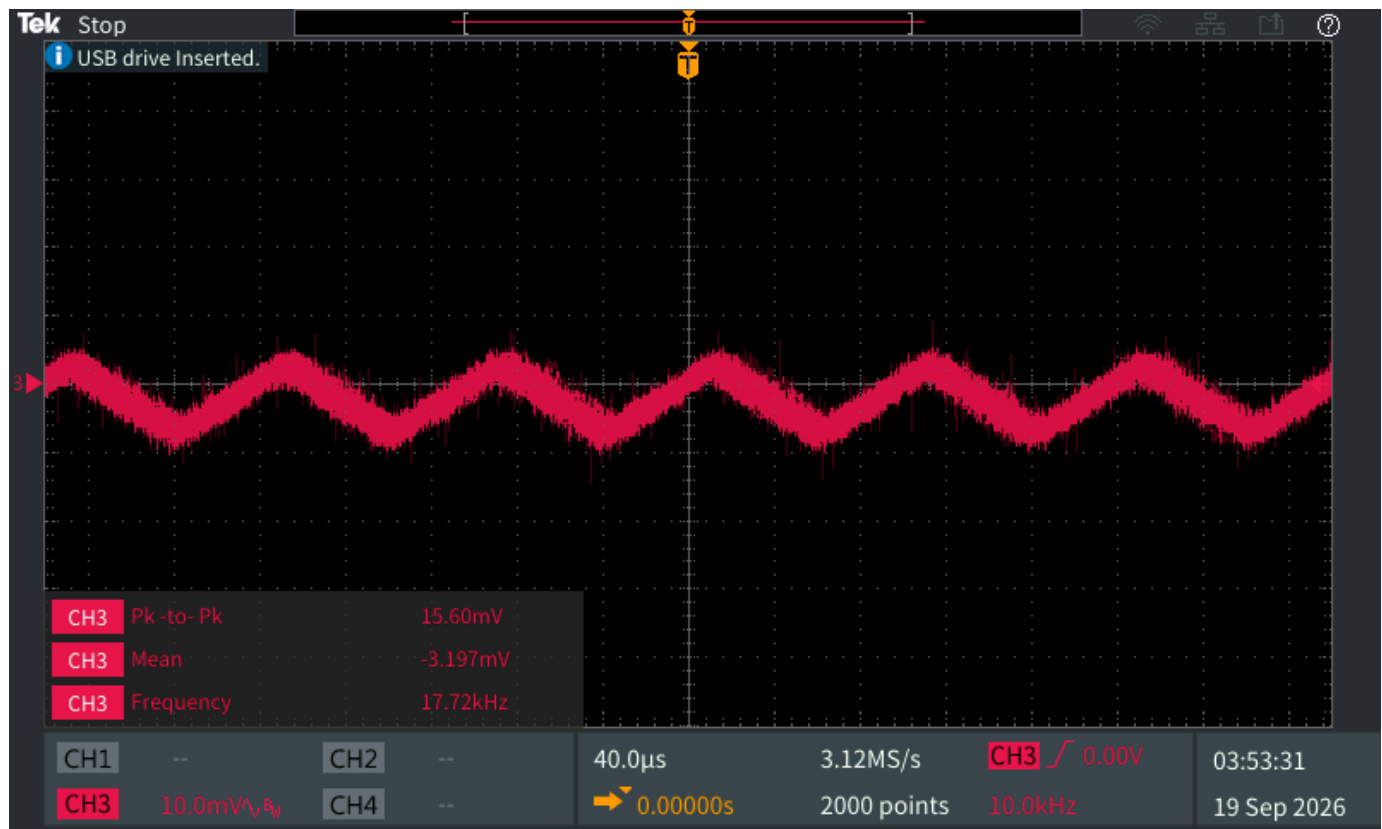


Fig. 21. Oscilloscope capture of the 10 kHz triangular wave with 10 mV amplitude, 2 V offset and AC coupling.

C. Frequency-Domain Captures and Measurements (Activity 3)

The team expected the FFT of the 10 kHz triangular wave to show a fundamental at 10 kHz and further components at integer multiples of it. The captured spectrum (Fig. 22) shows the fundamental at 10.0 kHz and the next component at 30.1 kHz.

After changing the signal to 20 kHz, the location of the main spectral line changed, as expected (Fig. 23). As expected, the pattern of the harmonics also changed completely when the signal was changed from a triangular wave to a sine wave (Fig. 24).

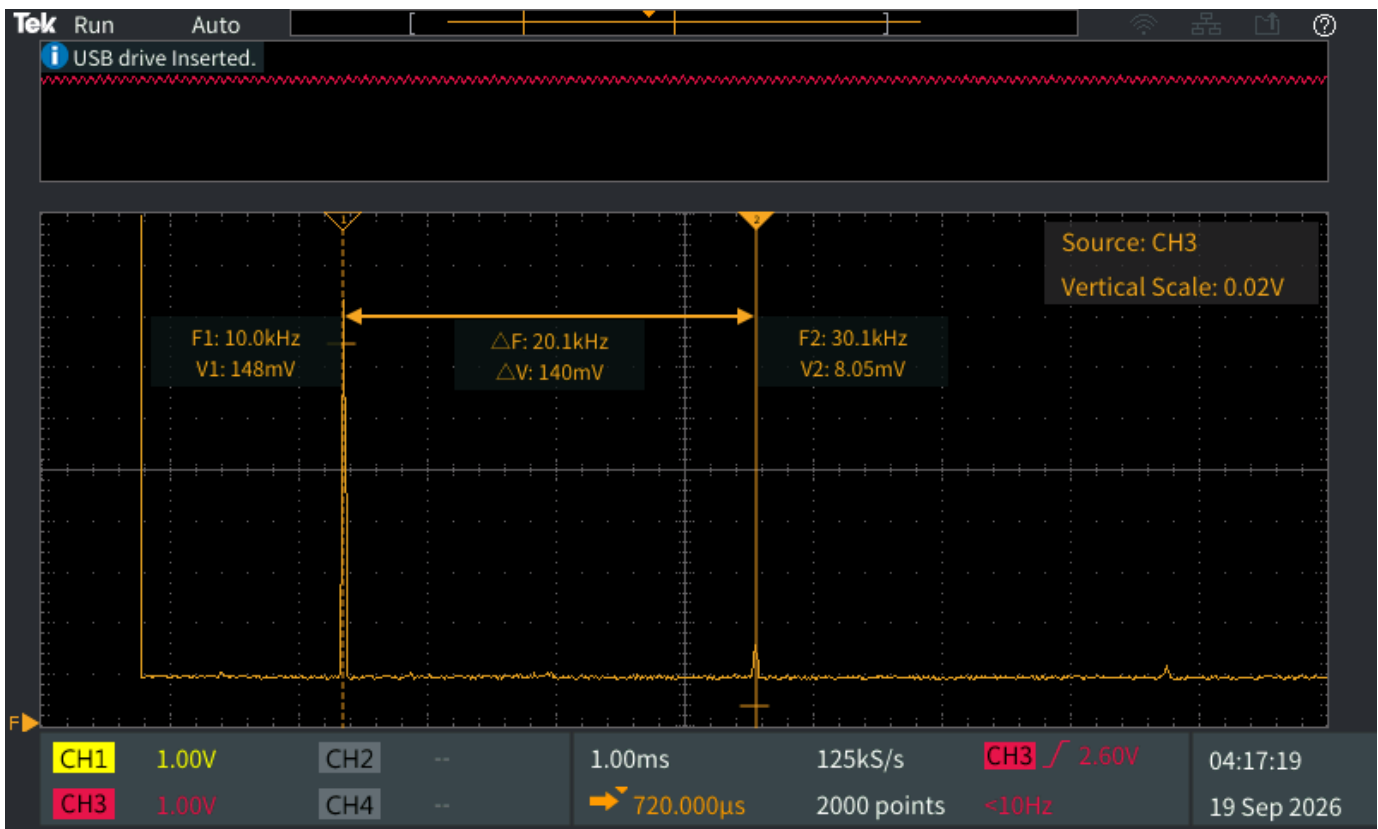


Fig. 22. FFT of the 10 kHz triangular wave.



Fig. 23. FFT of the 20 kHz triangular wave.

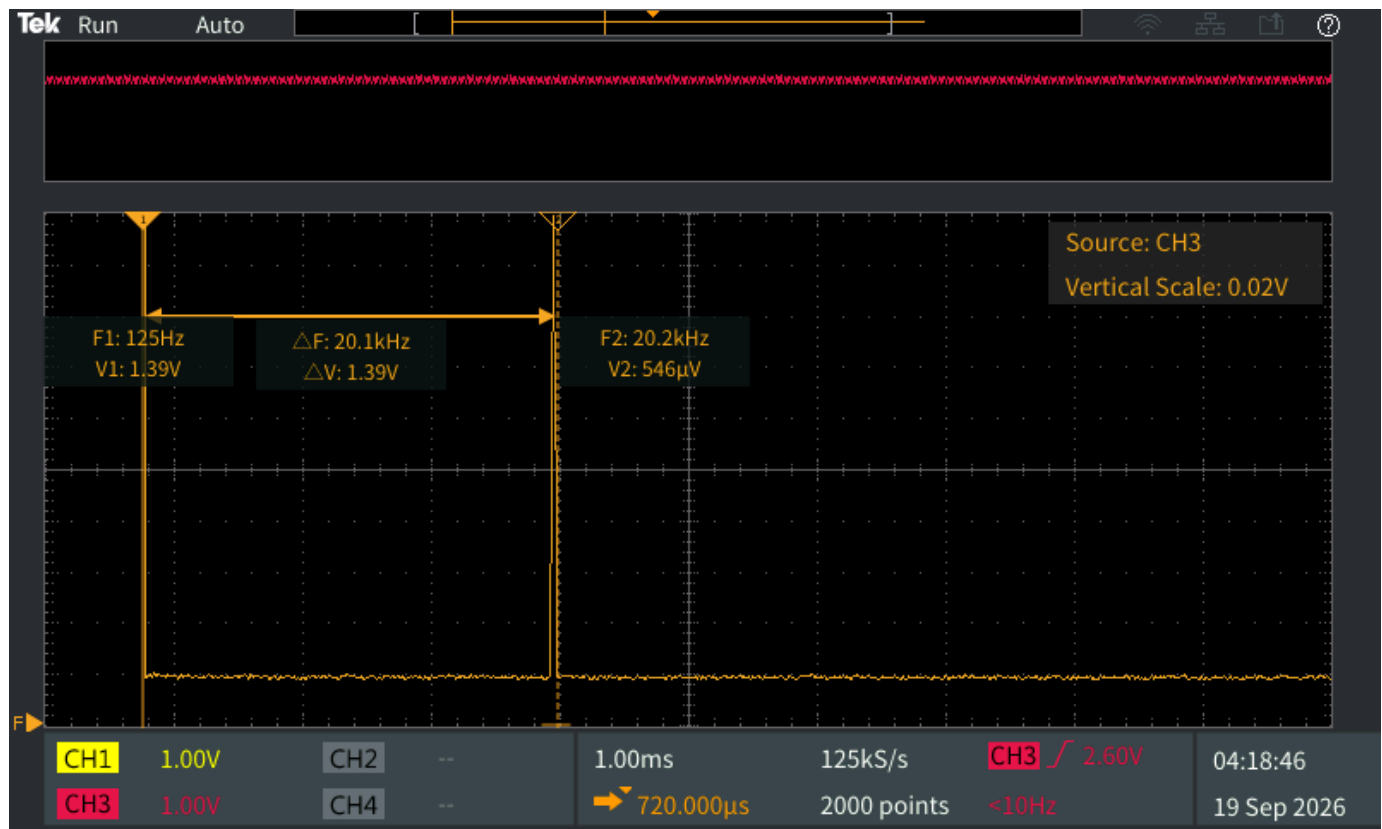


Fig. 24. FFT of the 20 kHz sine wave.

IV. CONCLUSION

This lab has given the team a better understanding of not only the instrumentation equipment and its calibration, but also of how simulation software processes this information. Spending a lab tinkering with the different oscilloscope functions has also contributed to a better overall understanding of a critical piece of equipment, relevant not only to the course but also to industry.

The biggest takeaways regarding the real-world versus simulation component of the course, in the opinion of this group, concern the Fast Fourier Transform. The team ran up against challenges during the lab with aliasing and with configuring the equipment to view proper waveforms. Encountering the same challenge again in the simulations allowed the group to target its weak points and overcome them. Each member has internalized the Shannon–Nyquist sampling theorem and has a greater understanding of its practical implications.

In simulation Activity 6, the triangle-wave harmonics were found to be very similar to those found in the final experiment of the lab. The team can therefore conclude that all systems are functional and, since the simulation matched reality, that the simulation can be considered a good double check.